# what is an API ?

API - Application Programming Interface.

www.pb.com




frontend backend

APIs are mechanism that enable two software components - such as the frontend and backend of an application to communicate with each other, using defined set of rules, protocols & data formats

frontend

Yatra.com

loc1 - Delhi

loc2 - mumbai

date - 4th nov

Submit

Communication (API)

IRCTC

Backend

def display_list_of_trains(loc1, loc2, date):

database

return list_of_trains

yatra.com — website

— android application

— ios application

— windows desktop applicat

— mac desktop applicatio

frontend

Backend

website - React, angular, django

mobile - kotlin
ios - swift
desktop - .dot




Python
Java
C/C++
javascript
etc

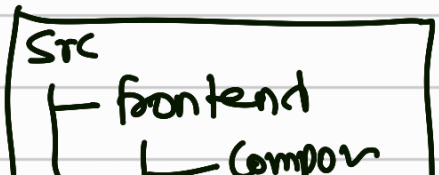
frontend → request → API → backend

frontend ← API ← Respose

without API — (monolithic Architecture)

Usecase- IRCTC company —

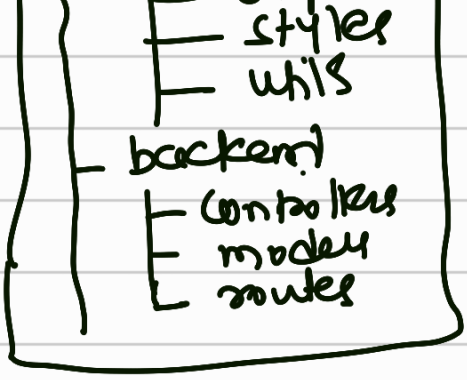
loc1 - delhi



loc2 - mumbai
date - 4th nov

Submit

o/p - list of trains




yatra  makemytrip  goibibo

locl -

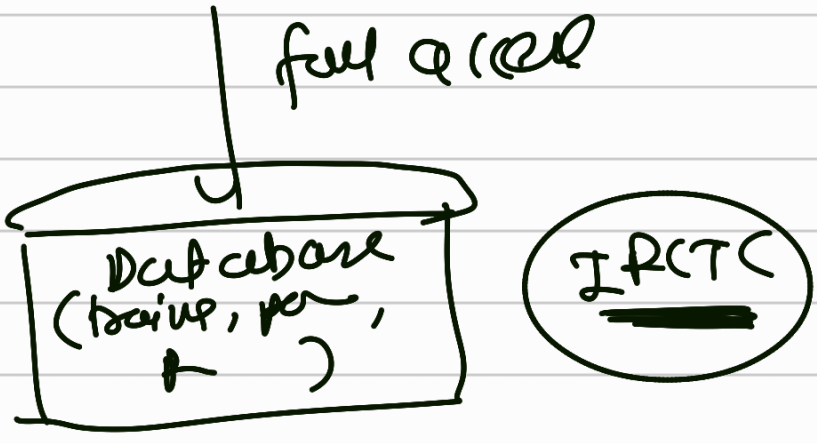
loc2 -

date -

Submit

option1 - Direct database access ✗

yatra / makemytr / goibib

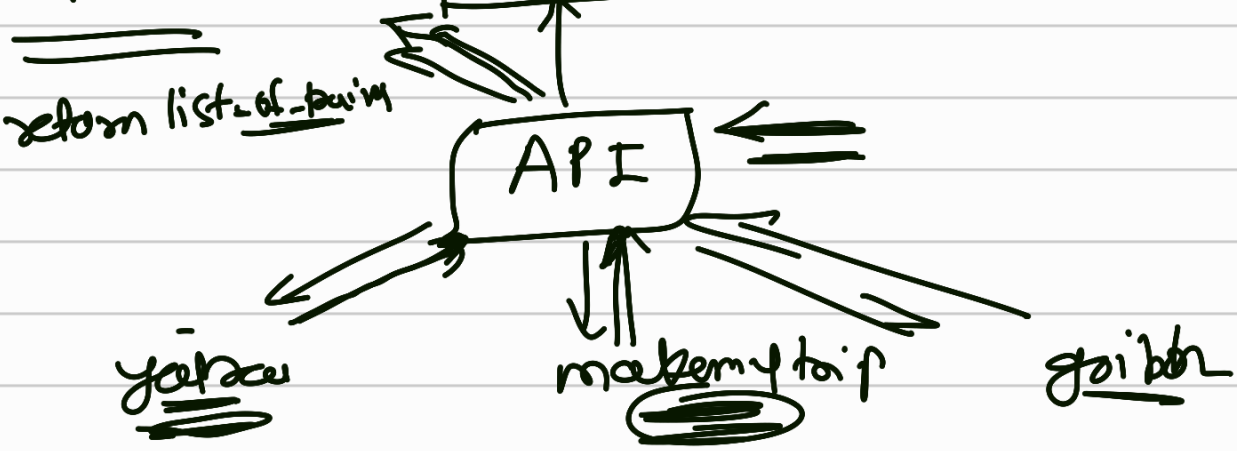



option2 -

def list_of_trains(loc1,
loc2, date):




IRCTC

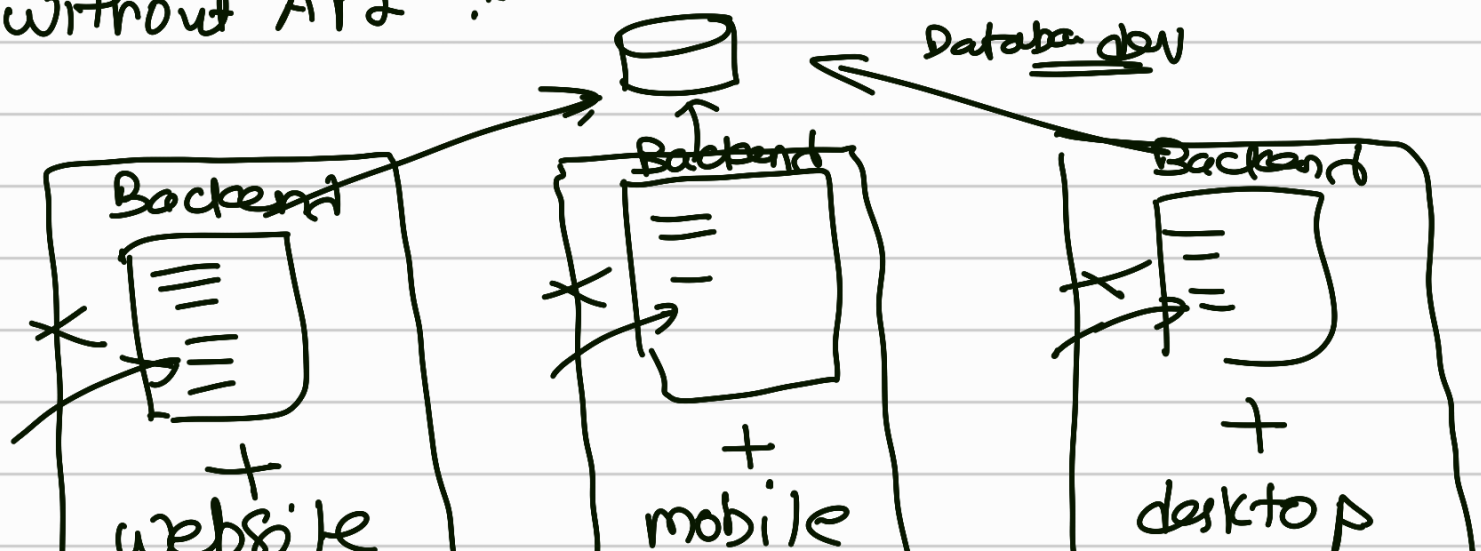


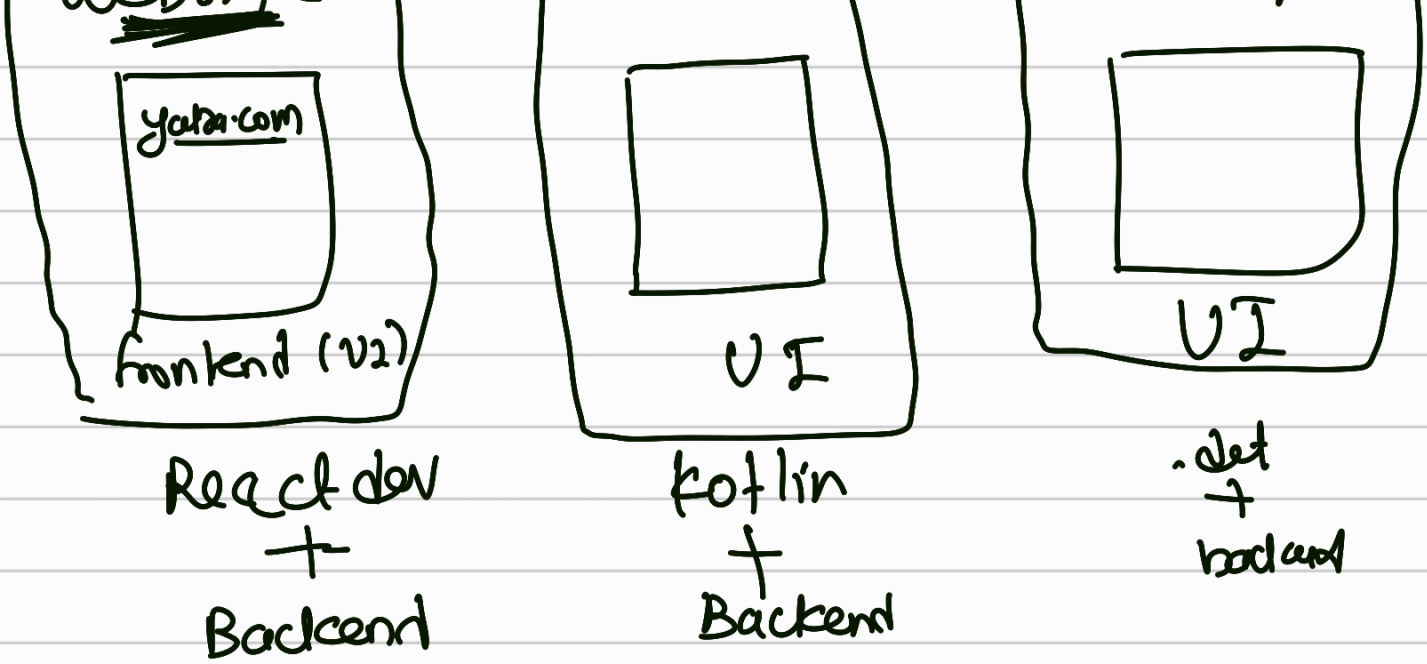

yaba.com/trains
yaba.com/flights

yaba.com/trains
def list_of_trains():

yaba.com/flight
def list_of_flight():

without API :-



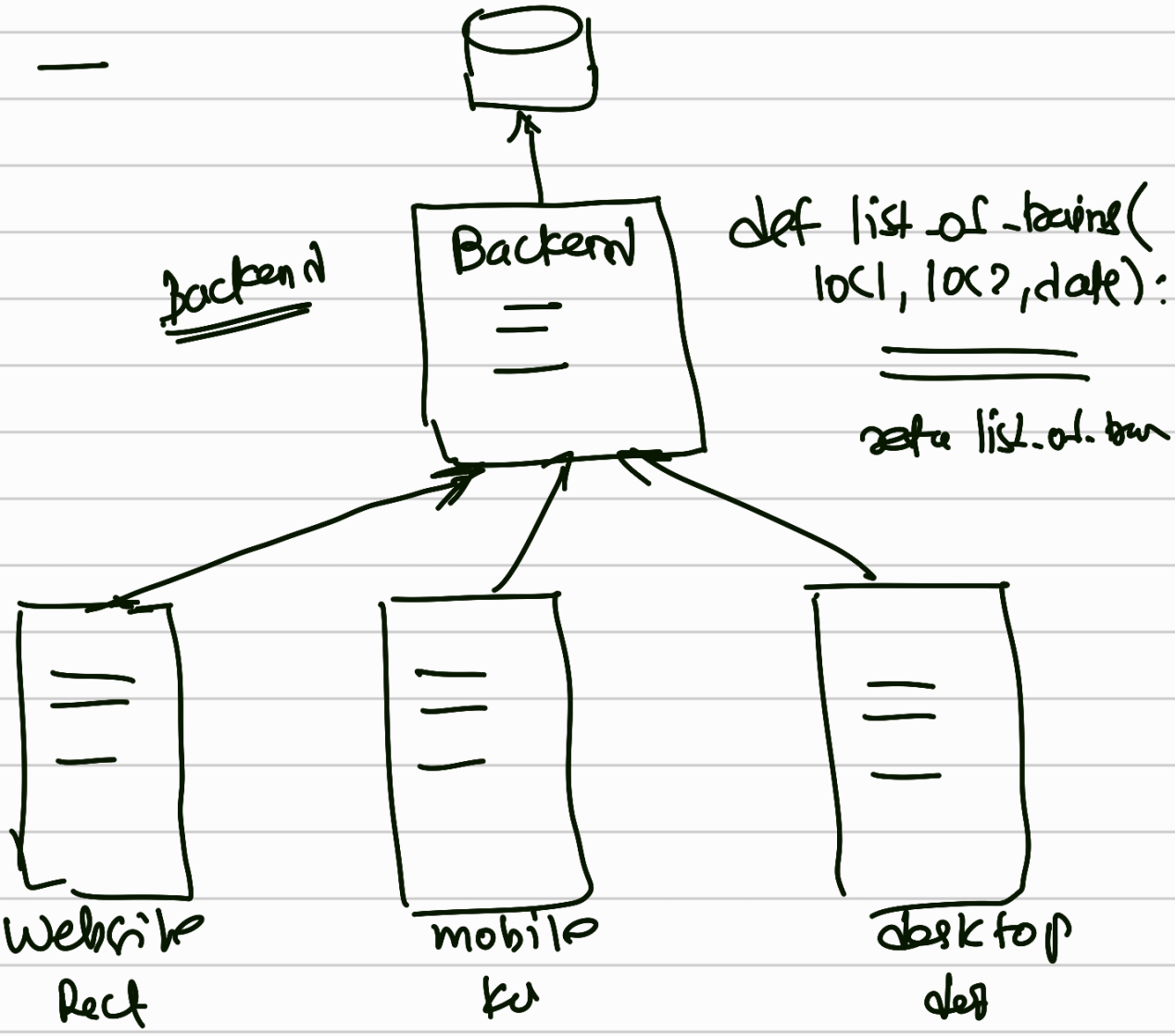



React dev
+
Backend

Kotlin
+
Backend

.dot
+
backend

API —




get (/trains)

def list_of_trains(loc1, loc2, date):

database connect

return list of trains between loc1 & loc2 for date

result = list_of_trains(delhi, mumbai, 4th nov)
print(result)

API → we can public our functions all over world

app.py

from fastapi import FastAPI

app = FastAPI()

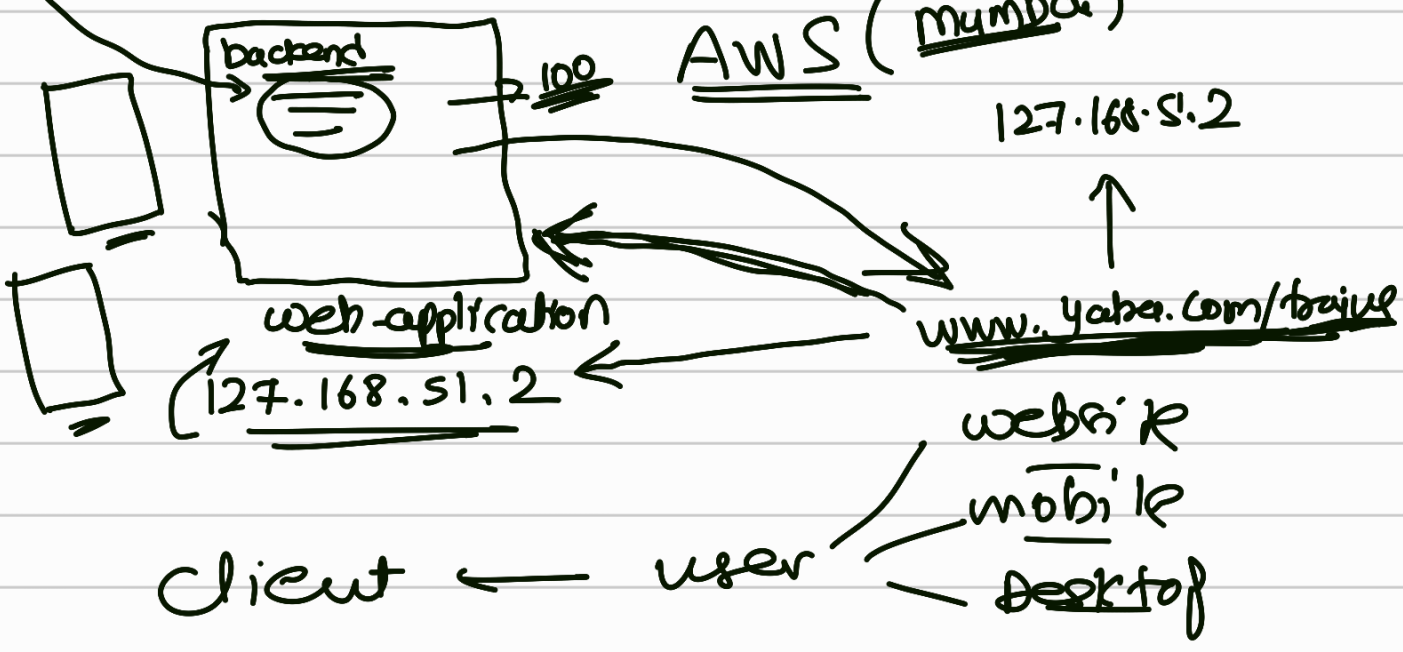
@app.get("/trains")

def get_list_of_trains(loc1, loc2, date):

access to database

return list of trains

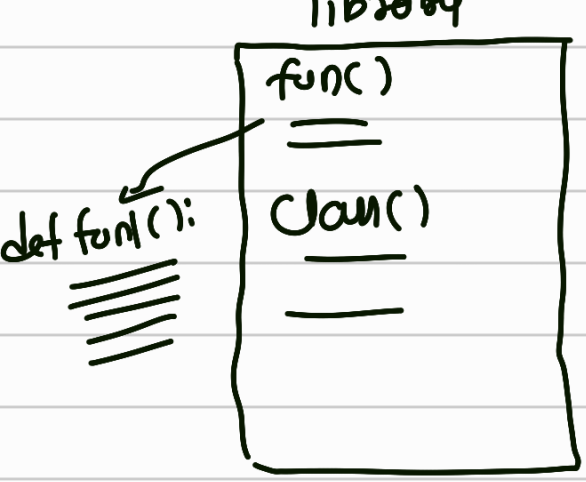
@app.get("/flights"):




DNS - Domain name system

What is fastAPI -

FastAPI is library of Python




len()

len(list1) → 5

from threading import Thread

class Student (Thread):

list1 = [1,2,3,4,5]

print (sum(list1))

o/p - 15

pip install fastapi

from fastapi import FastAPI

app = FastAPI()

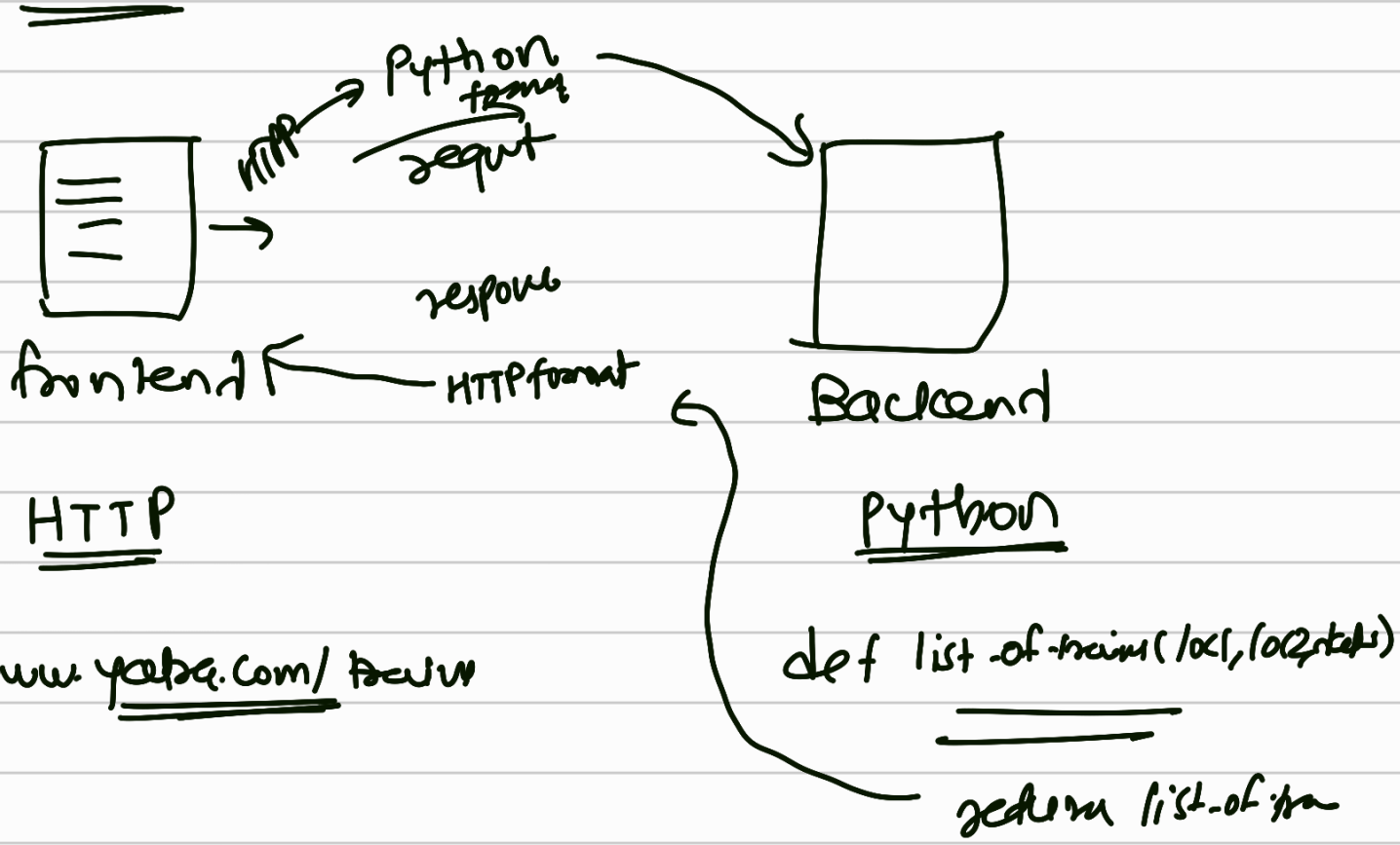
FastAPI is modern, high-performance, web framework, using this we can create APIs in python

Starlette

Pydantic

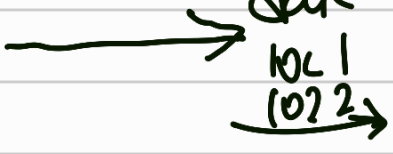
fastapi

Starlette –




Pydantic – Data validation

loc1 – str
loc2 – str
date – date




age – int

1 to 150 → ✗

-1 ✗

name – String

20 char ↑ ✗

3 char ↓ ✓